

Bash-Scripting assignment-02.

1. Create a bash script to **check if a directory is available or not**.

```
[root@ip-172-31-35-251 Neelimal]# vi check_directory
[root@ip-172-31-35-251 Neelimal]# chmod 777 check_directory
[root@ip-172-31-35-251 Neelimal]# cat check_directory
#!/bin/bash
#check directory present not available
#Date:20/12/25
#Modifications if any:
DIR="/root/Neelimal"
if [ -d "$DIR" ];
then
echo "Directory exists: $DIR"
else
echo "Directory does not exist: $DIR"
fi
```

Output:

```
[root@ip-172-31-35-251 Neelimal]# ./check_directory
Directory exists: /root/Neelimal
[root@ip-172-31-35-251 Neelimal]# |
```

2. Create a bash script to **create multiple files**.

```
[root@ip-172-31-35-251 Neelimal]# vi Create_multiple_files.sh
[root@ip-172-31-35-251 Neelimal]# chmod 777 Create_multiple_files.sh
[root@ip-172-31-35-251 Neelimal]# cat Create_multiple_files.sh
#!/bin/bash
#Create multiple lines
#Date:20/12/25
#Modifications if any

touch Neelimal2 Neelimal3 Neelimal4
echo "files are created Succesfully"
[root@ip-172-31-35-251 Neelimal]# ./Create_multiple_files.sh
files are created Succesfully
[root@ip-172-31-35-251 Neelimal]# |
```

3. Create a bash script to **take a backup of a directory**.

```
successful
[root@ip-172-31-35-251 Neelimal]# vi take_backup.sh
[root@ip-172-31-35-251 Neelimal]# chmod 777 take_backup.sh
[root@ip-172-31-35-251 Neelimal]# cat take_backup.sh
#!/bin/bash
#Take_Backup
#Date:20/12/25
#Modifications if any

read -p "Enter directory to backup: " Source_DIR
read -p "Enter backup location: " Backup_DIR

DATE=$(date +%F)
Backup_File="backup_$DATE.tar.gz"

mkdir -p "$Backup_DIR"
tar -czf "$Backup_DIR/$Backup_File" "$Source_DIR"
echo "successfull"
```

Bash-Scripting assignment-02.

Output:

```
[root@ip-172-31-35-251 Neelima1]# ./take_backup.sh
Enter directory to backup: /root/Neelima1
Enter backup location: /root/Neelima23
tar: Removing leading '/' from member names
successful
[root@ip-172-31-35-251 Neelima1]# |
```

4. Create a bash script to **install Nginx on an EC2 server.**

```
root@ip-172-31-35-251:~/Neelima1/scripts
[root@ip-172-31-35-251 scripts]# vi install_Nginx.sh
[root@ip-172-31-35-251 scripts]# chmod 777 install_Nginx.sh
[root@ip-172-31-35-251 scripts]# cat install_Nginx.sh
#!/bin/bash
#script to install Nginx
#Date:20/12/25
#Modifications if any
#check for root user
if [ "$(id -u)" -ne 0 ]
then
    echo "please run this script as root or using sudo"
exit 1
fi
echo "updating system packages..."
yum update -y

echo "installing nginx..."
yum install nginx -y

echo "enabling nginx on boot..."
systemctl enable nginx

echo "checking nginx status..."
systemctl status nginx --no-pager

echo "nginx installation completed successfully"
```

Step1: Connect to Ec2 instance

Create file of install nginx.sh as **vi install_Nginx.sh**

```
#!/bin/bash

#script to install Nginx

#Date:20/12/25

#Modifications if any

#check for root user

if [ "$(id -u)" -ne 0 ]

then

    echo "please run this script as root or using sudo"
```

Bash-Scripting assignment-02.

```
exit 1

fi

echo "updating system packages..."

yum update -y

echo "installing nginx..."

yum install nginx -y

echo "enabling nginx on boot..."

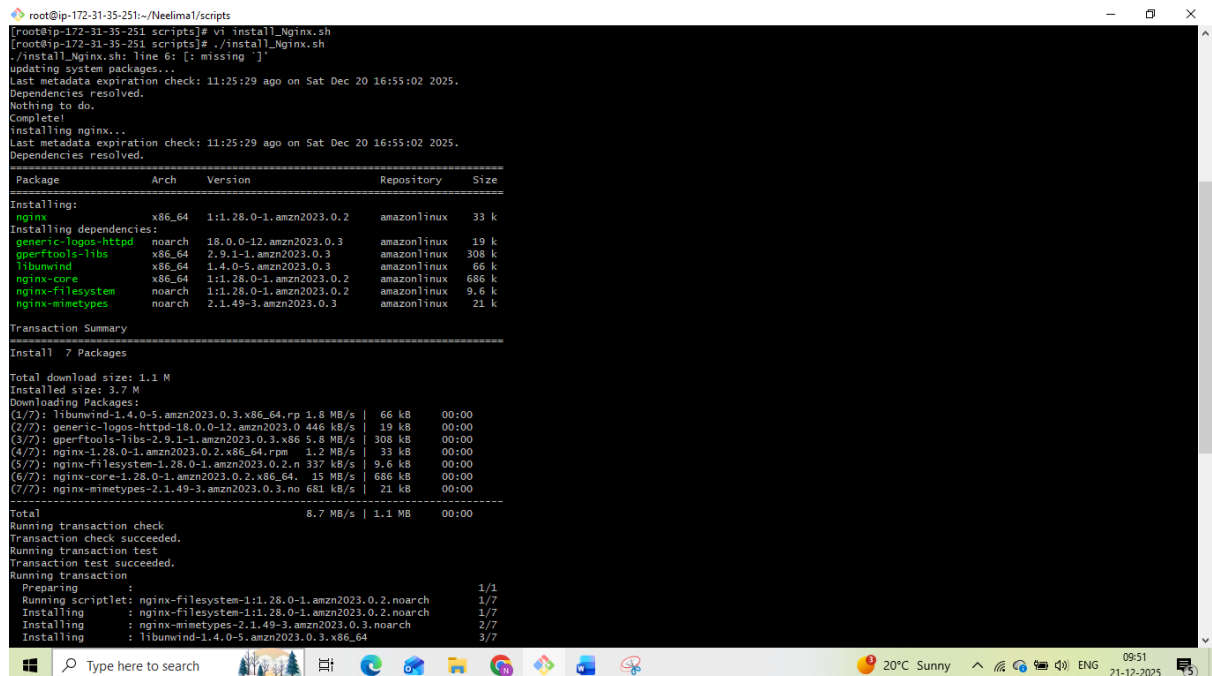
systemctl enable nginx

echo "checking nginx status..."

systemctl status nginx --no-pager

echo "nginx installation completed successfully"
```

- Save the file
- Give execution permission to Bash file
- Run the file as ./filename.sh



```
root@ip-172-31-35-251:~/Neelima/scripts
[root@ip-172-31-35-251 scripts]# vi install_Nginx.sh
[root@ip-172-31-35-251 scripts]# ./install_Nginx.sh
./install_Nginx.sh: line 6: [: missing `]'
updating system packages...
Last metadata expiration check: 11:25:29 ago on Sat Dec 20 16:55:02 2025.
Dependencies resolved.
Nothing to do.
Complete!
installing nginx...
Last metadata expiration check: 11:25:29 ago on Sat Dec 20 16:55:02 2025.
Dependencies resolved.

Package Arch Version Repository Size
-----
Installing:
nginx x86_64 1:1.28.0-1.amzn2023.0.2 amazonlinux 33 k
Installing dependencies:
generic-logos-httpd noarch 18.0-0-12.amzn2023.0.3 amazonlinux 19 k
gperftools-libs x86_64 2.9.1-1.amzn2023.0.3 amazonlinux 308 k
libumwind x86_64 1.4.0-5.amzn2023.0.3 amazonlinux 66 k
nginx-core x86_64 1:1.28.0-1.amzn2023.0.2 amazonlinux 686 k
nginx-filesystem noarch 1:1.28.0-1.amzn2023.0.2 amazonlinux 9.6 k
nginx-mimetypes noarch 2.1.49-3.amzn2023.0.3 amazonlinux 21 k

Transaction Summary
-----
Install 7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:
(1/7): libumwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.8 MB/s | 66 kB 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.446 kB/s | 19 kB 00:00
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 5.8 MB/s | 308 kB 00:00
(4/7): nginx-1.28.0-1.amzn2023.0.2.x86_64.rpm 1.2 MB/s | 33 kB 00:00
(5/7): nginx-filesystem-1:1.28.0-1.amzn2023.0.2.no 337 kB/s | 9.6 kB 00:00
(6/7): nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64. 15 MB/s | 686 kB 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.no 681 kB/s | 21 kB 00:00
-----
Total 8.7 MB/s | 1.1 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Running scriptlet: nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch 1/1
Installing : nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch 1/7
Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 2/7
Installing : libumwind-1.4.0-5.amzn2023.0.3.x86_64 3/7
```

Bash-Scripting assignment-02.

```
Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
gperf-tools-libs-2.9.1-1.amzn2023.0.3.x86_64
libunwind-1.4.0-5.amzn2023.0.3.x86_64
nginx-1:1.28.0-1.amzn2023.0.2.x86_64
nginx-core-1:1.28.0-1.amzn2023.0.2.x86_64
nginx-filesystem-1:1.28.0-1.amzn2023.0.2.noarch
nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
enabling nginx on boot...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr
/lib/systemd/system/nginx.service.
checking nginx status...
o nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: inactive (dead)
nginx installation completed successfully
[root@ip-172-31-35-251 scripts]#
```

5. Create a bash script to install Apache Tomcat on an EC2 server.

```
root@ip-172-31-35-251:~/Neelima1/scripts
[root@ip-172-31-35-251 scripts]# vi install_tomcat10.sh
[root@ip-172-31-35-251 scripts]# chmod 777 install_tomcat10.sh
[root@ip-172-31-35-251 scripts]# cat install_tomcat10.sh
#!/bin/bash
#script to install Apache Tomcat
#Date:20/12/25
#Modifications if any

TOMCAT_VERSION=10
TOMCAT_USER=tomcat
INSTALL_DIR=/opt/tomcat

#check root
if [ "$(id -u)" -ne 0 ];
then
    echo "Run as root or use sudo"
    exit 1
fi

echo "Installing Java (required for tomcat 10)..."
Yum install java-17-openjdk -y

echo "Creating tomcat user.."
useradd -r -m -U -d $INSTALL_DIR -s /bin/false $TOMCAT_USER

echo "Downloading Tomcat 10..."
cd /tmp
wget https://downloads.apache.org/tomcat/tomcat-10/v$TOMCAT_VERSION/bin/
apache-tomcat-$TOMCAT_VERSION.tar.gz

echo "Extracting tomcat.."
tar -xzf apache-tomcat-$TOMCAT_VERSION.tar.gz
mv apache-tomcat-$TOMCAT_VERSION/* $INSTALL_DIR/

echo "Setting permissions..."
chown -R $TOMCAT_USER:$TOMCAT_USER $INSTALL_DIR
chmod +x $INSTALL_DIR/bin/*.sh

echo "Starting Tomcat..."
sudo -u $TOMCAT_USER $INSTALL_DIR/bin/startup.sh
echo "Tomcat 10 installation completed successfully"
[root@ip-172-31-35-251 scripts]#
```

Bash-Scripting assignment-02.

Step1: Create ec2 instance and connect to instance.

Create file

```
#!/bin/bash
```

```
#script to install Apache Tomcat
```

```
#Date:20/12/25
```

```
#Modifications if any
```

```
TOMCAT_VERSION=10
```

```
TOMCAT_USER=tomcat
```

```
INSTALL_DIR=/opt/tomcat
```

```
#check root
```

```
if [ "$(id -u)" -ne 0 ];
```

```
then
```

```
    echo "Run as root or use sudo"
```

```
    exit 1
```

```
fi
```

```
echo "Installing Java (required for tomcat 10)..."
```

```
Yum install java-17-openjdk -y
```

```
echo "Creating tomcat user.."
```

```
useradd -r -m -U -d $INSTALL_DIR -s /bin/false $TOMCAT_USER
```

```
echo "Downloading Tomcat 10...."
```

```
cd /tmp
```

```
wget https://downloads.apache.org/tomcat/tomcat-
```

```
10/v$TOMCAT_VERSION/bin/apache-tomcat-$TOMCAT_VERSION.tar.gz
```

```
echo "Extracting tomcat.."
```

```
tar -xzf apache-tomcat-$TOMCAT_VERSION.tar.gz
```

```
mv apache-tomcat-$TOMCAT_VERSION/* $INSTALL_DIR/
```

```
echo "Setting permissions..."
```

```
chown -R $TOMCAT_USER:$TOMCAT_USER $INSTALL_DIR
```

```
chmod +x $INSTALL_DIR/bin/*.sh
```

```
echo "Starting Tomcat..."
```

```
sudo -u $TOMCAT_USER $INSTALL_DIR/bin/startup.sh
```

```
echo "Tomcat 10 installation completed successfully"
```

Bash-Scripting assignment-02.

```
[root@ip-172-31-35-251 scripts]# ./install_tomcat10.sh
Installing Java (required for tomcat 10)...
./install_tomcat10.sh: line 18: Yum: command not found
Creating tomcat user...
Downloading Tomcat 10...
--2025-12-21 04:45:28-- https://downloads.apache.org/tomcat/tomcat-10/v10/bin/apache-tomcat-10.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)[135.181.214.104]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-12-21 04:45:29 ERROR 404: Not Found.

Extracting tomcat..
tar (child): apache-tomcat-10.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
mv: cannot stat 'apache-tomcat-10/*': No such file or directory
Setting permissions...
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
Starting Tomcat...
sudo: /opt/tomcat/bin/startup.sh: command not found
Tomcat 10 installation completed successfully
[root@ip-172-31-35-251 scripts]#
```

6. Create a bash script to **check if the Nginx service is running**, if not running then script should start the service.

```
root@ip-172-31-35-251:~/Neelima1/scripts
[root@ip-172-31-35-251 scripts]# vi check_Nginx_running.sh
[root@ip-172-31-35-251 scripts]# chmod 777 check_Nginx_running.sh
[root@ip-172-31-35-251 scripts]# cat check_Nginx_running.sh
#!/bin/bash
#check if nginx is running or not
#Date:20/12/25
#Modification if any

SERVICE=nginx

if systemctl is-active --quiet $SERVICE;
then
    echo "nginx service is running"
else
    echo "nginx service is not running"
    echo "starting nginx service..."
    systemctl start $SERVICE

    #Recheck status
    if systemctl is-active --quiet $SERVICE;
    then
        echo "nginx service started successfully"
    else
        echo "Failed to start nginx service"
    fi
fi

[root@ip-172-31-35-251 scripts]#
```

Step 1: Create a instance and connect to instance

Step 2: Create a bash file to check nginx service is running or not.

```
#!/bin/bash
```

```
#check if nginx is running or not
```

```
#Date:20/12/25
```

```
#Modification if any
```

Bash-Scripting assignment-02.

```
SERVICE=nginx
if systemctl is-active --quiet $SERVICE;
then
    echo "nginx service is running"
else
    echo "nginx service is not running"
    echo "starting nginx service..."
    systemctl start $SERVICE

#Recheck status

if systemctl is-active --quiet $SERVICE;
then
    echo "nginx service started successfully"
else
    echo "Failed to start nginx service"
fi
fi
```

Output:

```
[root@ip-172-31-35-251 scripts]# ./check_Nginx_running.sh
nginx service is not running
starting nginx service...
nginx service started successfully
[root@ip-172-31-35-251 scripts]# |
```

Bash-Scripting assignment-02.

7. Create a bash script for a **calculator**.

```
[root@ip-172-31-35-251 Neelimal]# vi Simple_Calculator
[root@ip-172-31-35-251 Neelimal]# chmod 777 Simple_Calculator
[root@ip-172-31-35-251 Neelimal]# cat Simple_Calculator
#!/bin/bash
#Simple Calculator
#Date:20/12/25
#Modifications if any

echo "----- Simple Calculator -----"
echo "1. Addition"
echo "2. Subtraction"
echo "3. Multiplication"
echo "4. Division"

read -p "Enter your choice (1-4): " choice
read -p "Enter first number: " num1
read -p "Enter second number: " num2

case $choice in
    1)
        result=$((num1 + num2))
        echo "Result: $result"
        ;;
    2)
        result=$((num1 - num2))
        echo "Result: $result"
        ;;
    3)
        result=$((num1 * num2))
        echo "Result: $result"
        ;;
    4)
        if [ $num2 -eq 0 ]; then
            echo "Error: Division by zero not allowed"
        else
            result=$((num1 / num2))
            echo "Result: $result"
        fi
        ;;
    *)
        echo "Invalid choice"
        ;;
esac
```


Bash-Scripting assignment-02.

Output:

```
root@ip-172-31-35-251:~/Neelimal
[neelimal@ip-172-31-35-251 ~]$ ./Simple_Calculator
----- Simple Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Enter first number: 3
Enter second number: 2
Result: 5
[neelimal@ip-172-31-35-251 ~]$ ./Simple_Calculator
----- Simple Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 2
Enter first number: 3
Enter second number: 5
Result: -2
[neelimal@ip-172-31-35-251 ~]$ ./Simple_Calculator
----- Simple Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 3
Enter first number: 5
Enter second number: 14
Result: 70
[neelimal@ip-172-31-35-251 ~]$ ./Simple_Calculator
----- Simple Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 4
Enter first number: 1
Enter second number: 0
Error: Division by zero not allowed
[neelimal@ip-172-31-35-251 ~]$ ./Simple_Calculator
----- Simple Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 4
Enter first number: 3
Enter second number: 5
Result: 0
[neelimal@ip-172-31-35-251 ~]$ ./Simple_Calculator
----- Simple Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 5
Enter first number: 3
Enter second number: 4
Invalid choice
[neelimal@ip-172-31-35-251 ~]$
```

Bash-Scripting assignment-02.

8. Create a bash script to **check if a directory exists**, if not then create a directory.

```
root@ip-172-31-35-251:~/Neelima1
[root@ip-172-31-35-251 Neelima1]# vi check_Create_dir
[root@ip-172-31-35-251 Neelima1]# chmod 777 check_Create_dir
[root@ip-172-31-35-251 Neelima1]# cat check_Create_dir
#!/bin/bash
#Check or create a Directory
#Date:20/12/25
#Modifications if any

DIR="/root/Neelima1/scripts"
if [ -d "$DIR" ];
then
    echo "Directory already exists: $DIR"
else
    echo "Directory does not exist"
    echo "Creating directory..."
    mkdir -p "$DIR"

    if [ $? -eq 0 ];
    then
        echo "Directory created successfully: $DIR"
    else
        echo "Failed to create directory"
    fi
fi

[root@ip-172-31-35-251 Neelima1]# ./check_Create_dir
Directory already exists: /root/Neelima1/scripts
[root@ip-172-31-35-251 Neelima1]# |
```

9. Create a bash script to **delete the last 3 lines of a file**.

```
root@ip-172-31-35-251:~/Neelima1
[root@ip-172-31-35-251 Neelima1]# vi delete_lines
[root@ip-172-31-35-251 Neelima1]# vi devops_file
[root@ip-172-31-35-251 Neelima1]# vi delete_lines
[root@ip-172-31-35-251 Neelima1]# chmod +x delete_lines
[root@ip-172-31-35-251 Neelima1]# cat delete_lines
#!/bin/bash
#Delete last 3 lines in file
#Date:20/12/25
#Modifications if any:

FILE="devops_file"

#check if file exists

if [ -f "$FILE" ];
then
    sed -i '$d' "$FILE"
    sed -i '$d' "$FILE"
    sed -i '$d' "$FILE"
    echo "Last 3 lines deleted from $FILE"
else
    echo "File not found"
fi

[root@ip-172-31-35-251 Neelima1]# ./delete_lines
Last 3 lines deleted from devops_file
[root@ip-172-31-35-251 Neelima1]# |
```

Bash-Scripting assignment-02.

10. Bash script to monitor cpu and if it is more than 80% then send email notification.

```
[root@ip-172-31-35-251 Neelimal]# yum install mailx -y
Last metadata expiration check: 15:08:36 ago on Sat Dec 20 16:55:02 2025.
Dependencies resolved.
=====
Package           Architecture Version           Repository      Size
=====
Installing:
mailx             x86_64         12.5-43.amzn2023.0.1  amazonlinux    263 k
Transaction Summary
=====
Install 1 Package

Total download size: 263 k
Installed size: 497 k
Downloading Packages:
mailx-12.5-43.amzn2023.0.1.x86_64.rpm                1.0 MB/s | 263 kB    00:00
-----
Total                                               932 kB/s | 263 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Running scriptlet: mailx-12.5-43.amzn2023.0.1.x86_64 1/1
  Installing          : mailx-12.5-43.amzn2023.0.1.x86_64 1/1
  Running scriptlet: mailx-12.5-43.amzn2023.0.1.x86_64 1/1
  Verifying           : mailx-12.5-43.amzn2023.0.1.x86_64 1/1

Installed:
  mailx-12.5-43.amzn2023.0.1.x86_64

Complete!
```

```
[root@ip-172-31-35-251 Neelimal]# vi CPU_moni.sh
[root@ip-172-31-35-251 Neelimal]# chmod 777 CPU_moni.sh
[root@ip-172-31-35-251 Neelimal]# cat CPU_moni.sh
#!/bin/bash
#CPU Monitoring Script
#Date:20/12/25
#Modification if any

THRESHOLD=80
EMAIL="neelimarani1098@gmail.com"

CPU_USAGE=$(top -bn1 | grep "Cpu(s)" | awk '{print 100 - $8}')

CPU_USAGE=${CPU_USAGE%.*}

if [ "$CPU_USAGE" -gt "$THRESHOLD" ];
then
    echo "CPU usage is high: $CPU_USAGE%" | mail -s "CPU Alert on $(hostname)" "$EMAIL"
fi
[root@ip-172-31-35-251 Neelimal]# ./CPU_moni.sh
[root@ip-172-31-35-251 Neelimal]#
```

Bash-Scripting assignment-02.

11. Bash script to monitor disk space and if it is more than 80% then send email notification.

```
[root@ip-172-31-35-251 Neelimal]# vi Disk_Space.sh
[root@ip-172-31-35-251 Neelimal]# chmod 777 Disk_Space.sh
[root@ip-172-31-35-251 Neelimal]# cat Disk_Space.sh
#!/bin/bash
#Disk space Monitoring
#Date:20/12/25
#Modifications if any

THRESHOLD=80
EMAIL="neelimarani1098@gmail.com"

USAGE=$(df -h / |awk 'NR==2 {print $5}' | sed 's/%//')

if [ "$USAGE" -ge "$THRESHOLD" ]; then
    echo "warning: Disk usage is at ${USAGE}% ON $(hostname)" | \
        mail -s "Disk space Alert on $(hostname)" $EMAIL
fi

[root@ip-172-31-35-251 Neelimal]# ./Disk_Space.sh
[root@ip-172-31-35-251 Neelimal]#
```

12. Bash script to monitor memory and if it is more than 80% then send email notification.

```
root@ip-172-31-35-251:~/Neelimal
[root@ip-172-31-35-251 Neelimal]# vi Monitor_Memory
[root@ip-172-31-35-251 Neelimal]# chmod 777 Monitor_Memory
[root@ip-172-31-35-251 Neelimal]# cat Monitor_Memory
#!/bin/bash
#Monitor Memory usage
#Date:20/12/25
#Modification if any

THRESHOLD=80
EMAIL="neelimarani1098@gmail.com"

MEM_USAGE=$(Free | awk '/Mem/ {printf("%.0f"), $3/$2 * 100}')
if [ "$MEM_USAGE" -ge "$THRESHOLD" ]; then
    echo "warning: Memory usage is ${MEM_USAGE}% on $(hostname)" | \
        mail -s "Alert:High Memory usage" "$EMAIL"
fi

[root@ip-172-31-35-251 Neelimal]# ./Monitor_Memory
./Monitor_Memory: line 10: [: 70f: integer expression expected
[root@ip-172-31-35-251 Neelimal]# vi Monitor_Memory
[root@ip-172-31-35-251 Neelimal]# ./Monitor_Memory
[root@ip-172-31-35-251 Neelimal]#
```

Bash-Scripting assignment-02.

Exercise: Crontab Entries:

1. April 5th Midnight

```
root@ip-172-31-35-251:~  
[root@ip-172-31-35-251 ~]# vi /root/Neelimact.sh  
[root@ip-172-31-35-251 ~]# chmod 777 /root/Neelimact.sh  
[root@ip-172-31-35-251 ~]# vi /root/Neelimact.sh  
[root@ip-172-31-35-251 ~]# cat /root/Neelimact.sh  
#!/bin/bash  
#April 5th Midnight  
#Date:20/12/25  
#Modifications if any  
  
echo "April 5th Midnight" >> /root/Neelimact.sh  
[root@ip-172-31-35-251 ~]# crontab -e  
crontab: installing new crontab  
[root@ip-172-31-35-251 ~]# crontab -l  
0 0 5 4 * /root/Neelimact.sh  
  
[root@ip-172-31-35-251 ~]# |
```

* * * * *

| | | | |

| | | | └─ Day of Week

| | | └─── Month

| | └──── Day of Month

| └───── Hour

└────── Minute

Step 1: Create file by vi /root/Neelimact.sh

Step 2: Give executive permission to file

Step 3: Edit by command corntab -e and enter the fileds as per required

- Minutes Hours Date Months Week
- (0-59) (0-23)hrs (1-31)days (1-12)Months (0-6)

Bash-Scripting assignment-02.

2. 5th of Every November, January, June **if it is a Thursday**

Output :

crontab -e:- 0 0 5 1,6,11 4

```
[root@ip-172-31-35-251 ~]# vi /root/Neelimact1.sh
[root@ip-172-31-35-251 ~]# chmod 777 /root/Neelimact1.sh
[root@ip-172-31-35-251 ~]# cat /root/Neelimact1.sh
#!/bin/bash
#5th of Every November, January, June if it is a Thursday
#Date:20/12/25
#Modifications if any

echo "5th of Every November, January, June if it is a Thursday" /root/Neelimact1
.sh
[root@ip-172-31-35-251 ~]# corntab -e
-bash: corntab: command not found
[root@ip-172-31-35-251 ~]# crontab -e
crontab: installing new crontab
[root@ip-172-31-35-251 ~]# crontab -l
0 0 5 1,6,11 4 /root/Neelimact1.sh

[root@ip-172-31-35-251 ~]# |
```

3. At **05 and 27th minutes of 9, 10, 11 hours every day**

Output : 5,27 9,10,11 * * *

```
[root@ip-172-31-35-251 ~]# vi /root/Neelimact2.sh
[root@ip-172-31-35-251 ~]# chmod 777 /root/Neelimact2.sh
[root@ip-172-31-35-251 ~]# cat /root/Neelimact2.sh
#!/bin/bash
#05 and 27th minutes of 9, 10, 11 hours every day
#Date:20/12/25
#Modifications if any

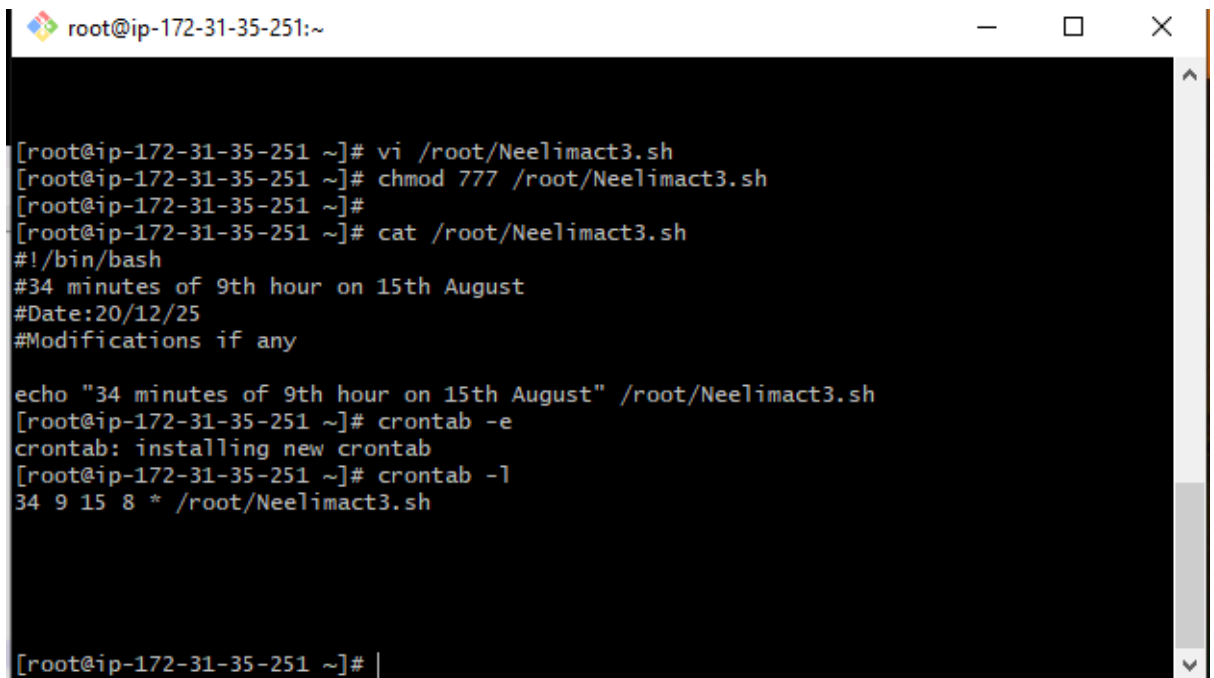
echo "05 and 27th minutes of 9, 10, 11 hours every day" /root/Neelimact2.sh
crontab: installing new crontabtab -e
[root@ip-172-31-35-251 ~]# crontab -l
5,27 9,10,11 * * * /root/Neelimact1.sh

[root@ip-172-31-35-251 ~]# |
```

Bash-Scripting assignment-02.

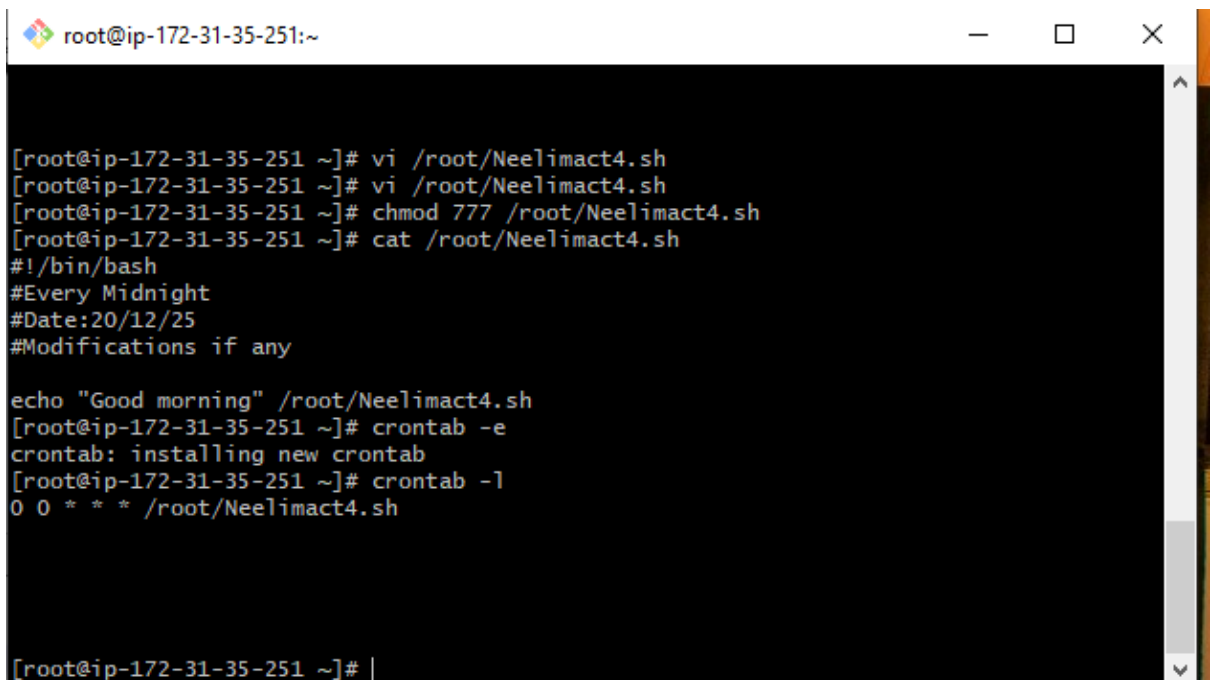
4. 34 minutes of 9th hour on 15th August

Output : 34 9 15 8 *



```
root@ip-172-31-35-251:~  
[root@ip-172-31-35-251 ~]# vi /root/Neelimact3.sh  
[root@ip-172-31-35-251 ~]# chmod 777 /root/Neelimact3.sh  
[root@ip-172-31-35-251 ~]#  
[root@ip-172-31-35-251 ~]# cat /root/Neelimact3.sh  
#!/bin/bash  
#34 minutes of 9th hour on 15th August  
#Date:20/12/25  
#Modifications if any  
  
echo "34 minutes of 9th hour on 15th August" /root/Neelimact3.sh  
[root@ip-172-31-35-251 ~]# crontab -e  
crontab: installing new crontab  
[root@ip-172-31-35-251 ~]# crontab -l  
34 9 15 8 * /root/Neelimact3.sh  
  
[root@ip-172-31-35-251 ~]# |
```

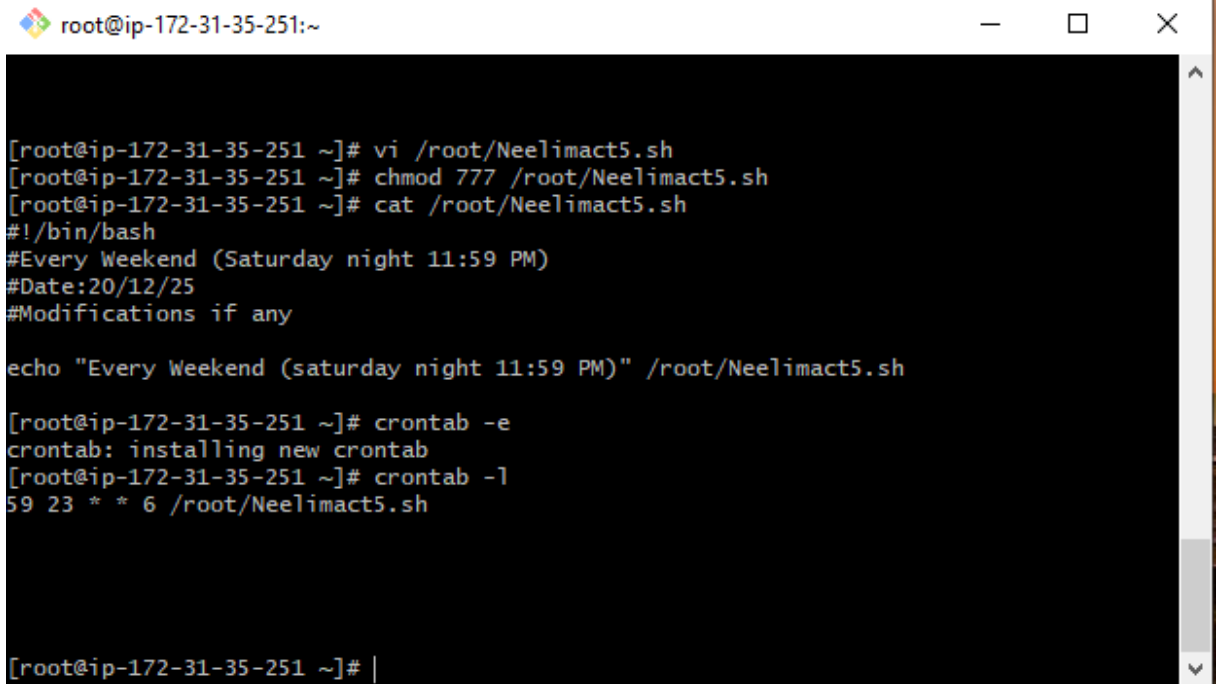
5. Every midnight



```
root@ip-172-31-35-251:~  
[root@ip-172-31-35-251 ~]# vi /root/Neelimact4.sh  
[root@ip-172-31-35-251 ~]# vi /root/Neelimact4.sh  
[root@ip-172-31-35-251 ~]# chmod 777 /root/Neelimact4.sh  
[root@ip-172-31-35-251 ~]# cat /root/Neelimact4.sh  
#!/bin/bash  
#Every Midnight  
#Date:20/12/25  
#Modifications if any  
  
echo "Good morning" /root/Neelimact4.sh  
[root@ip-172-31-35-251 ~]# crontab -e  
crontab: installing new crontab  
[root@ip-172-31-35-251 ~]# crontab -l  
0 0 * * * /root/Neelimact4.sh  
  
[root@ip-172-31-35-251 ~]# |
```

Bash-Scripting assignment-02.

6. Every Weekend (Saturday night 11:59 PM)



```
root@ip-172-31-35-251:~  
[root@ip-172-31-35-251 ~]# vi /root/Neelimact5.sh  
[root@ip-172-31-35-251 ~]# chmod 777 /root/Neelimact5.sh  
[root@ip-172-31-35-251 ~]# cat /root/Neelimact5.sh  
#!/bin/bash  
#Every Weekend (Saturday night 11:59 PM)  
#Date:20/12/25  
#Modifications if any  
  
echo "Every Weekend (saturday night 11:59 PM)" /root/Neelimact5.sh  
  
[root@ip-172-31-35-251 ~]# crontab -e  
crontab: installing new crontab  
[root@ip-172-31-35-251 ~]# crontab -l  
59 23 * * 6 /root/Neelimact5.sh  
  
[root@ip-172-31-35-251 ~]# |
```

7. After every reboot

@reboot /root/startup.sh

- @reboot → runs **once each time the system boots**
- No time/date fields are required
- The command runs when **crond starts**
- chmod +x /root/startup.sh